

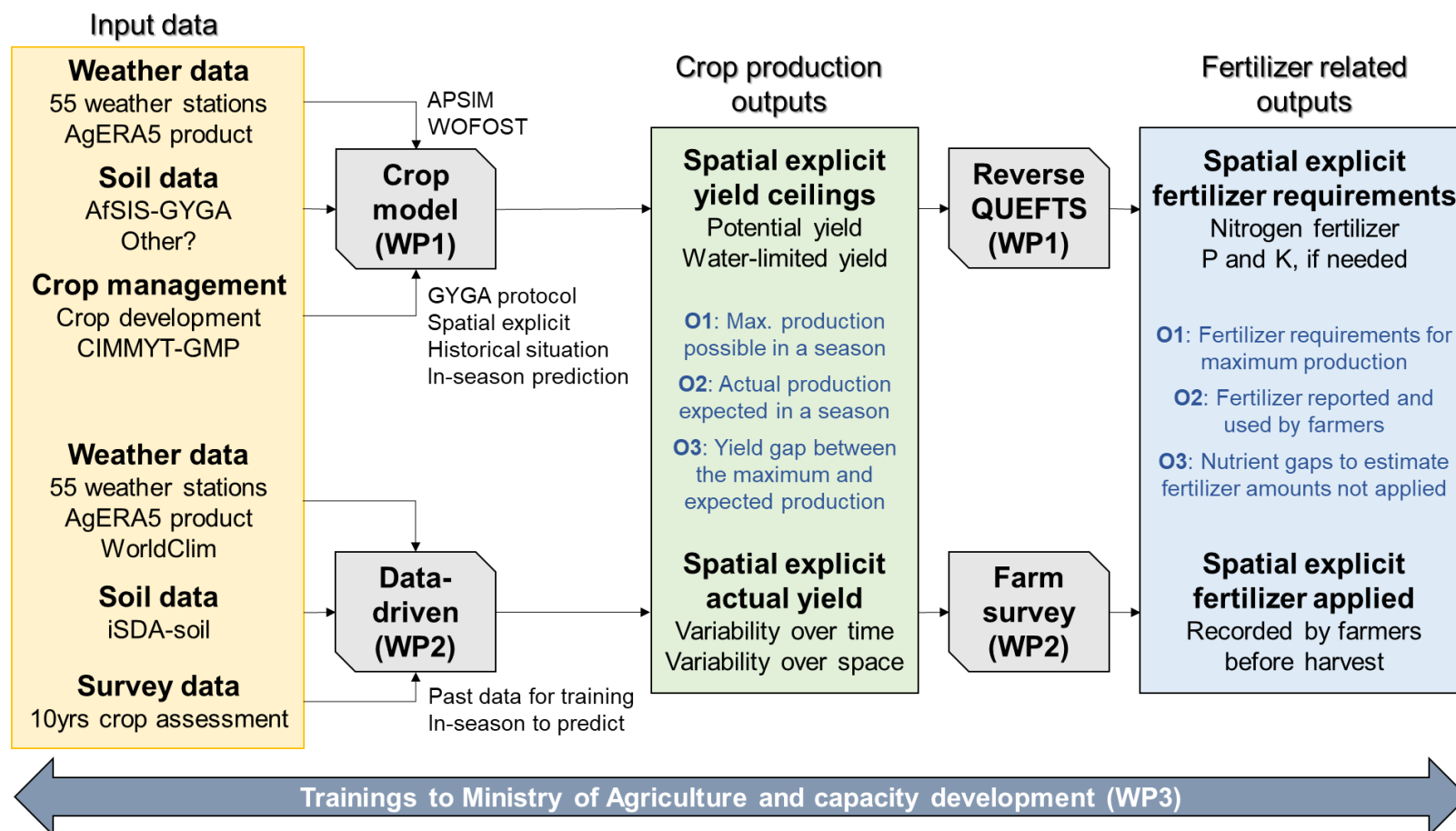


Figure 1. Components of the early warning system for crop production, and related fertilizer requirements, to be hosted at the Ministry of Agriculture in Zimbabwe. The early warning system is organized into three work packages (WP): WP1 deals with crop modelling for estimation of yield ceilings and respective nutrient requirements; WP2 deals with data-driven approaches for in-season yield prediction of crop yields under on-farm conditions; finally, WP3 deals with training and capacity development of the future users and maintainers of the early warning system. Key outputs are indicated as O1, O2, and O3 and these are to be assembled into a real-time data visualization dashboard containing all input and output data.

Establishment of an early warning hub to monitor food security in Zimbabwe

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1. Background

Agriculture remains Zimbabwe's most direct route to reducing poverty, hunger, and malnutrition, given that 67% of the population resides in rural areas and relies on smallholder farming as a source of livelihood. Of the 72% of people living in poverty, 84% are also in rural areas. Ensuring food security at the household and national level is a key objective for both government and development partners. In Zimbabwe, where over 70% of the population depends on agriculture for livelihood collecting agriculture data as an early warning tool becomes critical. For an early warning system to be effective and functional, it must be done timely and must be accurate. The main outputs of the evaluations, which include crop and livestock production estimates, are used as an early warning tool for programming and planning by the Government and partners to address food security and nutrition issues. A stable country depends on the status of food security. Agriculture production must be derived from reliable data collection methods to obtain reliable food security status.

The current early warning system used in Zimbabwe to monitor food security and fertilizer requirements needs more in-depth technical support and state-of-the-art agronomic modelling tools and datasets to improve early warning for early action across the country. Digital agriculture is the seamless integration of digital technologies into crop and livestock management and other processes in agriculture. Beyond technical tools, there is also a high demand for training and capacity development to ensure the Ministry of Agriculture is able to uptake an improved early warning system as part of the early warning hub it is currently putting into place. The proposed early warning hub aims to make available real-time, relevant, quality, harmonised and cost-efficient agriculture statistical data and information, which includes: (1) crop area and yield estimation for various crops; (2) livestock identification and traceability; (3) agroecological crop zoning; (4) land information systems; (5) information on water and dams; and (6) agricultural trade information for sector planning and formulation, implementation, monitoring and evaluation of policies and programmes. The prototype to be developed in this project is envisaged to be a platform integrating critical data for monitoring food security. Thus, it will create a legacy upon which future investments on it can be built upon with minimum cost and effort.

2. Partnership

This project aims to address the demand of the Ministry of Agriculture in Zimbabwe to improve the existing early warning system for food security and fertilizer requirements in the country. As such, the following partnership is envisaged to answer to this demand: (a) CIMMYT will be the lead coordinating partner and developer of the early warning system, (b) Ministry of Agriculture of Zimbabwe will guide the development of the early warning system to ensure it answers the needs of the government, (c) University of Zimbabwe will help with the development of specific components of the early warning system through support for a MPhil student, and (d) and advanced university/research center will provide technical backstopping and support during the development of key components of the early warning system in WP1 and WP2. A web designer will be involved in WP3 through a consultancy

contract to create the user interface and background infrastructure of the digital atlas, where input data, outputs, and workflows will be repositied.

3. Simulation of water-limited yield and related nutrient requirements (WP1)

WP1 focus on the simulation of crop-specific potential and water-limited yields, i.e., the maximum yield possible under irrigated or rainfed conditions, respectively, and associated fertiliser requirements, using crop models (Figure 1). To do so, we will leverage upon different sources of weather, soil, and crop management data needed to run crop growth simulation models. Two sources of weather data will be used and compared namely weather station data from the Government of Zimbabwe and AgERA5, a global database with weather data for crop modelling purposes. Daily records of rainfall, minimum, and maximum temperature over a period of 30 years will be provided by the Government of Zimbabwe for 55 weather stations across the country, which is considered the 'gold standard'. The soil data needed for crop modelling will be retrieved from AfSIS-GYGA, a continental-wide database with spatially predicted soil water properties for sub-Saharan Africa. Finally, crop management data on sowing, anthesis, and harvest/maturity dates needed for model calibration will be obtained from extension officers working with the Ministry of Agriculture and cross-validated against variety trials conducted by CIMMYT across the country or existing secondary data.

Two protocols will be compared to simulate the potential and water-limited yield so that results are spatial explicit. First, the widely used protocol of the Global Yield Gap Atlas (see www.yieldgap.org) will be followed. This protocol consists of a prior delineation of weather station buffer zones for the climate zones containing a given share of crop harvested area. Subsequently, soil types and cropping systems are identified for each weather station buffer zone and crop models are run for the respective weather station used to delineate the buffer zones. In this way, potential and water-limited yields are simulated using primary weather data according to a protocol that allows up-scaling the results to the regional level. The crop model WOFOST will be used to make the simulations for the Global Yield Gap Atlas. The second protocol consists of running the crop model using the spatial explicit input data on weather, soil, and crop management data. Although this approach is coarser, it relies on widely available weather data making the transferability of the protocol to other regions straightforward. The crop model APSIM will be used to conduct the spatial explicit crop model simulations. WOFOST is the crop model powering the early warning system for food security in Europe (see MARS system website), whereas APSIM has been widely used as decision support tool in dryland, rainfed, farming systems of Australia (see example).

Crop model simulations will be first run for historical weather data following the two protocols explained above. Crop model simulations will be subsequently conducted with the same protocols for in-season prediction of the water-limited yield, using weather forecasts provided by the Ministry of Agriculture of Zimbabwe.

Lastly, once potential and water-limited yields are simulated, the minimum N requirements required to reach that yield level will be estimated using a 'reverse QUEFTS' approach. QUEFTS is a simple rule-based model to estimate crop yield given the amount of fertilizer applied, or to estimate the amount of fertilizer needed to reach a particular yield.

The outputs of WP1 will be hosted at www.yieldgap.org and/or at the Ministry of Agriculture of Zimbabwe (see WP3) and include:

- 1) Database of weather, crop management, and yield data for maize across Zimbabwe.

- 2) Spatial explicit potential and water-limited yield for maize in Zimbabwe, and associated estimates of potential maize production across administrative and agro-ecological zones of the country.
- 3) Spatial explicit nutrient requirements to reach potential and water-limited yield at national level and per administrative and agro-ecological zone.

4. Data-driven prediction of crop yield and on-farm conditions (WP2)

WP2 will focus on the analysis of actual yield (i.e., the yield observed in farmers' fields) variability and its prediction in time and space. This WP will leverage upon the crop assessment survey conducted by the Ministry of Agriculture over the past 10 yrs, comprising detailed management and yield data more than 50.000 farms each year. The survey data will be coupled with spatial explicit secondary data on weather (see WP1 as well as other data sources such as WorldClim) and soils (mostly the 30m resolution iSDA-soil) to characterize the biophysical conditions of each farm.

Data-driven models will be developed based on the crop assessment survey. First, a yield gap decomposition will be conducted to identify the key limiting factors to maize production across the main production areas of Zimbabwe as well as for different types of farms. The yield gap of rainfed crops refers to the difference between the water-limited yield (estimated in WP1) and the actual yield observed on farmers' fields. Stochastic frontier analysis will be combined with agronomic knowledge to decompose the yield gap into efficiency, resource, and technology yield gaps. Secondly, machine learning models will be developed to explain and predict yield variability across farms. Explanatory models will be used to identify the key drivers of actual yield variability, which serve as a validation of the results of the stochastic frontier analysis. Predictive models will be developed to predict actual yields over time and space, while identifying the variables influencing yield prediction most and assessing model performance based on different statistical metrics. In-season actual yield predictions will also be conducted in addition to actual yield predictions for different years and spatial units. The crop assessment survey is conducted in two separate rounds: one round before harvest where management data and the expected actual yield are recorded, and a second round where the actual yield is measured on-farm. Therefore, data for the first round of the crop assessment survey will be used as input to a data-driven model trained on historical data to predict yield at the end of the growing season, which will be further validated with the yield measured at harvest time. In-season yield predictions will be automated so that yield data are predicted as soon as the first round of the crop assessment survey is conducted.

Special attention will be paid to records of fertilizer amounts and field areas so that accurate estimates of fertilizer application rates on farmers' fields can be obtained. These observed fertilizer rates will be compared to the fertilizer requirements to narrow the yield gap (as estimated in WP1) so that 'nutrient gaps' (i.e., the nutrient deficit needed to realize the potential crop yield in a given location) can be calculated.

Outputs of WP2 include:

- 1) Spatial explicit data on actual yield and yield gaps, as well as fertiliser use and 'nutrient gaps', for maize in Zimbabwe. Estimates of maize production at national and sub-national level will also be provided.
- 2) Identification of the key limiting factors to maize yield gaps across the country, and according to agro-ecological zone and types of farms.

- 3) Validated data-driven methods to explain and predict actual yield variability over space and time.
- 4) ODK forms for data collection integrated with data-driven methods for real time yield predictions based on the first round of the crop assessment survey.

5. Training and capacity development activities (WP3)

WP3 will have a strong focus on training and capacity development to ensure staff of the Ministry of Agriculture in Zimbabwe can reproduce, maintain, and improve the prototype of the early warning system developed in WP1 and WP2 (Figure 1). Such activities will entail monthly meetings where scripts developed for WP1 and WP2 will be presented, discussed, and made available to the partners along with a session on capacity development in the programming tools needed to execute those scripts.

As the project will produce numerous workflows and scripts and handle diverse datasets invaluable for future research and development activities, we propose to reposit all these materials in a data visualization dashboard to be hosted at the Ministry of Agriculture in Zimbabwe. Beyond repositing data and scripts, the outputs will be presented in the form of a digital '*Atlas of agricultural production and farming systems in Zimbabwe*' so that these can be used for decision making at policy level and for targeting activities of non-governmental organizations working on agriculture in Zimbabwe.

The key outputs of WP3 thus focus on ensuring training of local partners in Zimbabwe and repositing of all data and tools in ways that can ensure their future usability. These include:

- 1) Fully documented and reproducible workflows and scripts for the different analysis undertaken in WP1 and WP2.
- 2) Data visualization dashboard compiling the outputs of WP1 and WP2 in the format of a digital '*Atlas of agricultural production and farming systems in Zimbabwe*' to be hosted at the Ministry of Agriculture.
- 3) Monthly trainings on data science between CIMMYT, University of Zimbabwe, and Ministry of Agriculture of Zimbabwe. External partners will be invited once or twice during the project lifetime.

6. Contribution to National Determined Contributions (NDCs)

In Zimbabwe, attaining the objectives of Vision 2030 will be guided by the National Development Strategy that aims to achieve an upper-middle-income society by 2030. The overall goals are informed by Vision 2030, the Malabo Declaration of 2014, and the Sustainable Development Goals (SDGs) of 2015. To achieve food security, the country aims to have robust early warning systems for early action on food security and related natural resource management. Accurate data and reliable agricultural data and statistical information are required to: (1) inform public policy and evaluate agricultural policy and ongoing agriculture development initiatives, which include, among others, the Agriculture Recovery Plan, the Livestock Recovery and Growth Plan, and the Horticulture Recovery and Growth Plan; (2) feed into and underpin the planning processes of agriculture investment initiatives such as irrigation revitalisation, rehabilitation and development, and farm mechanisation; (3) compilation of National Accounts; and (4) facilitating investment and trade decisions including prices and trade information for trade agreement negotiations.

The establishment of an early warning system will facilitate the real-time collection, collation, analysis, synthesis, and visualization of agriculture statistical data and information, raise awareness of the critical role of accurate information for planning, policy research, monitoring and evaluation while promoting the wider adoption and use of evidence-based decision-making, while assisting in season decisions by farmers, government, and other stakeholders. To achieve, the Ministry of Agriculture is investing in an early warning hub. The ultimate impact of the early warning hub will be improved agriculture development outcomes and strengthened governance of the National Agriculture Statistical Systems while enhancing monitoring and evaluation of the Agriculture and Food Systems Transformation Strategy.

7. Next steps to extend the early warning system

The aim of this project is to develop a prototype of an early warning system for maize, the main food security crop cultivated in Zimbabwe, based on state-of-the-art methods and protocols for yield gap analysis and yield prediction over time and space. Yet, few other activities could be incorporated to refine/validate the prototype or to extend it to consider other features relevant for policy making. Thus, depending on budget availability, the following activities could be considered:

1. Expand of the early warning system to other crops beyond maize. The methods and approaches proposed are generic, so such expansion depends on the availability of reliable farm data.
2. Ex-ante impact assessment of future climate change impacts on crop production. Projected weather data for different RCPs are globally available and could be used as input in the crop models and in the data-driven models.
3. Analysis of a network of existing field experiments under optimal growth conditions to (a) evaluate the performance of the crop models, and (b) disentangle the contribution of G x E x M interactions to historical maize yields in Zimbabwe.
4. On-farm validation of agronomic options to intensify maize production in Zimbabwe's smallholder sector (e.g., climate responsive management), as informed by the yield gap analysis.
5. Exploration of intensification pathways for selected farming systems to unravel on which types of farms and farm sizes yield gap closure can contribute to improved livelihoods in addition to satisfying food security at national level.